The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 July 2004.

INCH-POUND

MIL-PRF-19500/429J
7 April 2004
SUPERSEDING
MIL-PRF-19500/429H
30 December 2002

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, FAST RECOVERY, TYPES 1N5615, 1N5617, 1N5619, 1N5621, 1N5623, 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, 1N5623UL, 1N5615US, 1N5617US, 1N5619US, 1N5621US, 1N5623US, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon, fast recovery power rectifier diodes that are hermetic glass encapsulated. Four levels of product assurance are provided for each encapsulated device as specified in MIL-PRF-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figure 1 (similar to DO-7), figures 2 and 3 surface mount (D-5A), figures 4 and 5 (JANHC and JANKC).

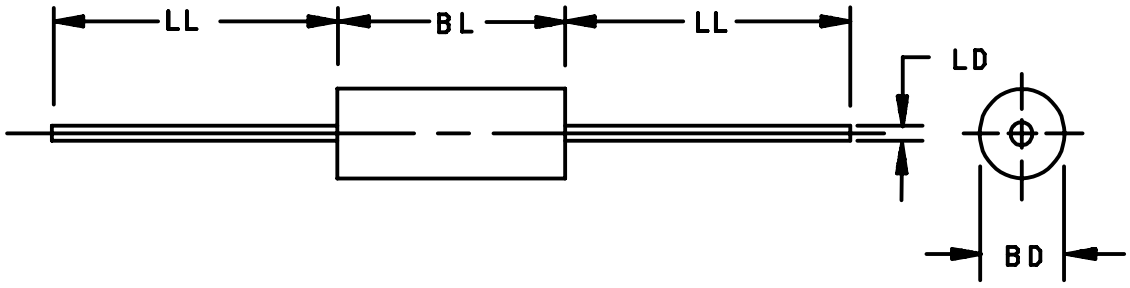
1.3 Maximum ratings. Unless otherwise specified T_C = 25°C.

Col. 1	Col. 2	Col. 3	Col. 4		Col. 5	Col. 6	Col. 7	Col. 8	Col. 9	Col. 10
Types (1)	V_R	V_{RWM}	I_O		t_{rr}	T_{STG} T_J	I_{FSM} T_A = +100°C I_O = 750 mA dc t_P = 8.3 ms	Barometric pressure (reduced)	$R_{\theta JL}$ $R_{\theta JEC}$	$Z_{\theta JX}$
			T_A = +55°C (2) (3)	T_A = +100°C (3) (4)						
	V dc	V(pk)	A dc	mA dc	ns	°C	A(pk)	mmHg	(5)	°C/W
1N5615	200	200	1	750	150	-65	25	N/A		4.5
1N5617	400	400	1	750	150		25	8		4.5
1N5619	600	600	1	750	250	to	25	8		4.5
1N5621	800	800	1	750	300		25	33		4.5
1N5623	1,000	1,000	1	750	500	+175	25	33		4.5

(1) Electrical characteristics for UL and US suffix are identical to the corresponding non-suffix device.
(2) From 1 A dc at T_A = +55°C, to .75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
(3) For the 1 A dc rating at +55°C and the 750 mA dc rating at +100°C, no special mounting, heat sinking, or forced-air flow across exposed areas of the device is required.
(4) From .75 A dc at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
(5) $R_{\theta JL} \leq$ 38°C/W at L = .375 inch (9.53 mm); for UL and US suffix type devices. $R_{\theta JEC} \leq$ 7.0°C/W.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43216-5000, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://www.dodssp.daps.mil.

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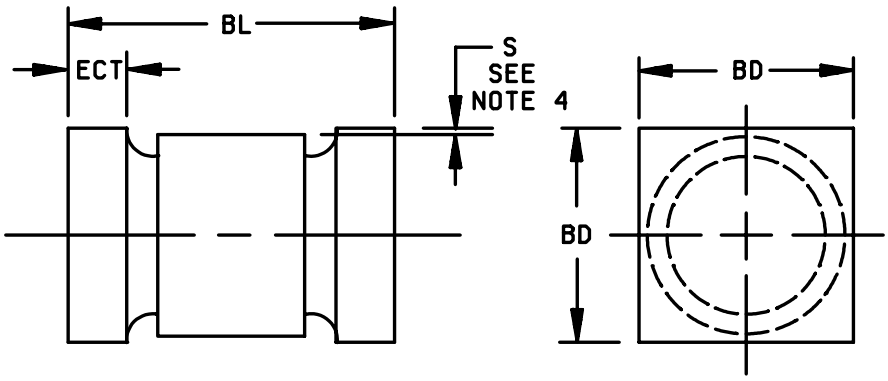


Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	.065	.110	1.65	2.79	3
LD	.026	.033	.66	.84	
BL	.130	.225	3.30	5.71	4
LL	1.00	1.30	25.40	33.02	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimension BD shall be measured at the largest diameter.
4. Dimension BL shall include all uncontrolled areas of the device.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

* FIGURE 1. Physical dimensions (for non-UL and non-US suffix devices only) (similar to DO-7).

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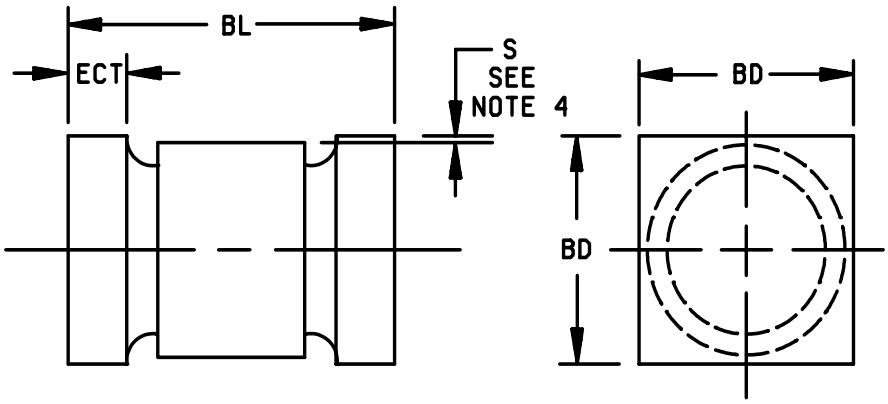


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.225	4.27	5.72
ECT	.019	.028	0.48	0.71
S	.003		0.08	
BD	.091	.125	2.31	3.18

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for types 1N5615UL, 1N5617UL, 1N5619UL, 1N5621UL, and 1N5623UL, (surface mount devices) (D-5A).

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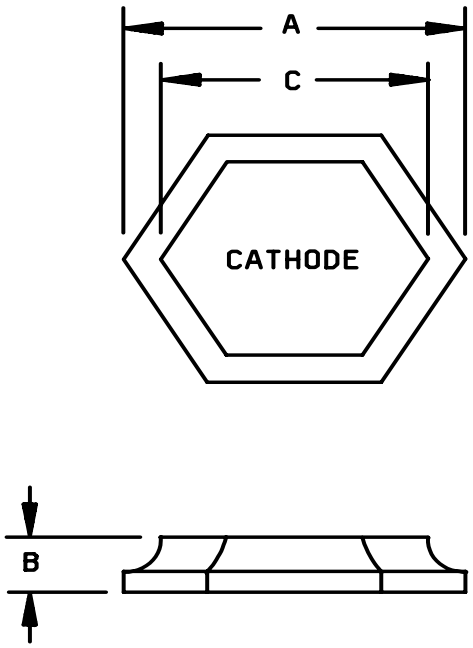


Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.168	.200	4.27	5.08
ECT	.019	.028	0.48	0.71
S	.003		0.08	
BD	.091	.103	2.31	2.62

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Dimensions are pre-solder dip.
4. Minimum clearance of glass body to mounting surface on all orientations.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 3. Physical dimensions for types 1N5615US, 1N5617US, 1N5619US, 1N5621US, and 1N5623US, (surface mount devices) (D-5A).

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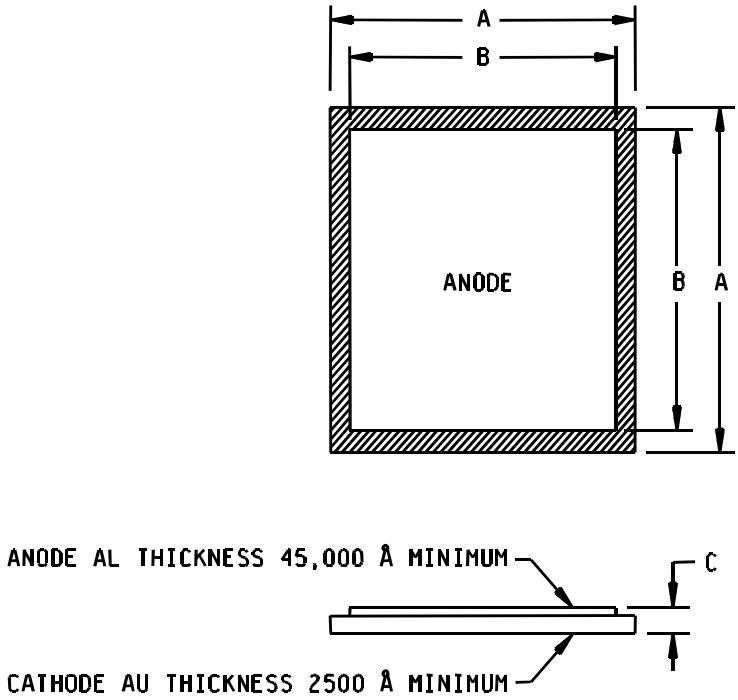
A - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Gold 10,000 Å minimum.
   Back metal: Gold 4,000 Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions, JANHCA and JANKCA die.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. The physical characteristics of the die are:
   Top metal: Aluminum (anode) 45,000 Å minimum.
   Back metal: Gold (cathode) 2,500 Å minimum.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 5. Physical dimensions, JANHCB and JANKCB die.

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

* DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

* DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.dap.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

* 3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows:

EC	End cap.
UL	Long-body unleaded or surface mounted diodes (square end-caps).
US	Short-body unleaded or surface mounted diodes (square end-caps).

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (similar to DO-7), figures 2 and 3 surface mount (D-5A), figures 4 and 5 (JANHC and JANKC).

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3.4.1 Lead finish. Unless otherwise specified, lead or end cap finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. When solder alloy is used for finish the maximum lead temperature is limited to 175°C maximum. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Diode construction. These devices shall be constructed in a manner and using material which enable the diodes to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5.1 Encapsulant material. In addition to those categories of hermetically sealed package requirements specified in MIL-PRF-19500, fused-metal-oxide to metal shall also be acceptable.

3.6 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6.1 Marking of UL and US version. For UL and US version only, all marking (see 3.6) may be omitted from the body, but shall be retained on the initial container.

3.6.2 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting color band may be used.

3.7 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E qualification shall be performed herein for qualification or requalification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein shall be performed by the first inspection lot to this revision to maintain qualification.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with appendix G of MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTXV and JANTX level
1a 1b	Required Required	Not required Required (JANTXV only)
2	Not required	Not required
3a (1) 3c	Required Thermal impedance (see 4.3.1 and 4.4.1)	Required Thermal impedance (see 4.3.1 and 4.4.1)
4	Not applicable	Not applicable
5	Not applicable	Not applicable
6	Not applicable	Not applicable
7a 7b	Not applicable Optional	Not applicable Optional
8	Required	Not required
9	Required I_{R1} and V_{FM1}	Not required
10	Method 1038 of MIL-STD-750, condition A	Method 1038 of MIL-STD-750, condition A
11	I_{R1} and V_F, $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc	I_{R1} and V_F
12	Required, see 4.3.2	Required, see 4.3.2
(2) 13	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).	Subgroup 2 of table I herein: $\Delta I_{R1} \leq 100$ percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq \pm 0.1$ V dc. Scope display evaluation (see 4.5.4).
14a (3) 14b	Not applicable Required	Not applicable Required
15	Required	Not required
16	Required	Not required

(1) Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-PRF-19500, screen 3 prior to this thermal test.

(2) $Z_{\theta JX}$ is not required in screen 13, if already previously performed.

(3) For clear glass diodes, the hermetic seal (gross leak) may be performed at any time after temperature cycling.

4.3.1 Thermal impedance. Thermal impedance $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750. The maximum limit for $Z_{\theta JX}$ in screening (table IV of MIL-PRF-19500) shall be derived by each vendor by means of statistical process control and applied in screening of all subsequent lots. This limit shall not exceed the table I, subgroup 2 limit. See 4.4.1 for test conditions.

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4.3.1.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with method 3101 of MIL-STD-750 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

* 4.3.2 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.3, 4.5.3.1) adjust I_O to achieve the required T_J.

4.3.3 Screening (JANHC and JANKC). Screening of die shall be in accordance with appendix G of MIL-PRF-19500. As a minimum, die shall be 100-percent probed to ensure compliance with table I, subgroup 2. Burn-in duration for the JANKC level follows JANS requirements; the JANHC follows JANTX requirements.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein. If alternate screening is being performed in accordance with E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied in accordance with 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in table I, subgroup 2 inspection:

a. I_H..... 10 A minimum.

b. t_H..... 10 ms.

c. I_M ..... 10 mA.

d. t_{MD} ..... 100 ∝s maximum.

The maximum limit for $Z_{\theta JX}$ in table I, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.

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* 4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500. For B5, if a failure occurs, resubmission shall be at the test conditions of the original sample.

	Subgroup	Method	Condition
	B3	1056	0°C to +100°C, 25 cycles.
	B3	1051	-55°C to +175°C, 100 cycles.
	B3	4066	I_{FSM} = rated I_{FSM} (see col. 7 of 1.3); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 0, V_{RWM} = 0.
	B4	1037	$I_O = I_{O2}$ rated minimum (see col. 4 of 1.3); V_R = rated V_{RWM} (see col. 3 of 1.3, and 4.5.3); 2,000 cycles.
	B5	1027	$I_O = I_{O2}$ rated minimum (see col. 4 of 1.3); apply V_R = rated V_{RWM} (see col. 3 of 1.3 and 4.5.3) adjust or I_O to achieve T_J minimum; f = 50-60 Hz; n = 45, c = 0.
			Option 1: T_A = + 30°C max. ; T_J = 225°C minimum; t = 216 hours.
		or	Option 2: T_A = + 30°C max. ; T_J = 200°C minimum; t = 1,000 hours.
*	B6	3101	$R_{\theta JL}$ (maximum) ≤ 308°C/W; L = .375 inch (9.53 mm).
		or 4081	For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
*	B7		Peak reverse power, see 4.5.6 and figure 6 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan n = 10, c = 0, end-points, see 4.4.2.

4.4.2.2 Group B inspection, table VIb (JAN, JANTX and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	1056	0°C to +100°C, 10 cycles.
B2	1051	-55°C to +175°C, 25 cycles.
B3	1027	$I_O = I_O$ rated (see col. 4 of 1.3) minimum; adjust I_O to achieve the required T_J apply V_R = rated V_{RWM} (see col. 3 of 1.3), f = 50-60 Hz (see 4.5.3.1).
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein. See table III for delta limits when applicable.

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* 4.4.3.1 Group C inspection, table VII of MIL-PRF-19500.

	Subgroup	Method	Condition
*	C2	1056	0°C to +100°C, 10 cycles.
*	C2	1051	-55°C to +175°C, 25 cycles.
*	C2	2036	Tension: Test condition A; weight = 12 pounds; t = 30 seconds. Lead fatigue: Test condition E; weight 1 pound. NOTE: Both tension and lead fatigue are not applicable for US devices.
*	* C5	3101 or 4081	See 4.5.5.
	C6	1026	I_O = I_O rated (rated see col. 4 of 1.3) minimum; adjust I_O to achieve the required T_J apply V_R = rated V_{RWM} (see col. 3 of 1.3), f = 50-60 Hz (see 4.5.3.1).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein; except, $Z_{\theta JX}$ need not be performed. See table III for delta limits when applicable.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

* 4.5.3.1 Free air burn-in. Deliberate heat sinking, baffles to create an oven, forced air-cooling or heating is prohibited unless otherwise approved by the qualifying activity. The use of a current limiting or ballast resistor is permitted provided that each DUT still sees the full P_t (minimum) and that the minimum applied voltage, where applicable, is maintained through out the burn-in period. T_J = 135°C minimum for screening and T_J = 150°C for 4.4.2 and 4.4.3 life tests. Use method 3100 of MIL-STD-750 to measure T_J.

4.5.4 Scope display evaluation. Scope display evaluation shall be stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in table I, subgroup 4 shall be performed on a scope. Reverse current (I_{BR}) over the knee shall be 500 ∝A peak.

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4.5.5 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with test method 3101 or 4081 of MIL-STD-750. Read and record data in accordance with table II herein and shall be included in the qualification report. Forced moving air or draft shall not be permitted across the devices during test. The maximum limit under these test condition shall be $R_{\theta JL} \leq 38°C/W$ for L = .375 (9.53 mm). $R_{JEC} = 7°C/W$ for UL or US devices, L = 0.

a. I_H = 5 A dc minimum.

b. $t_{H =}$ thermal equilibrium.

c. I_M = 1.0 mA to 10 mA.

d. t_{MD} = 100 ∝s maximum.

The device shall be allowed to reach equilibrium at current I_H before the measurement shall be made ($t_H \geq 25$ sec).

LS = Lead spacing = .375 inch (9.53) mm minimum for leaded devices and LS = 0 minimum for unleaded devices as defined (see figure 7) (Metric equivalents are given for general information only):

4.5.6 Peak reverse power test. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current. A 20 microsecond square waveform may also be used with the approval of the qualifying activity (see figure 6).

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
	2071					
Subgroup 1 Visual and mechanical inspection						
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc (pulsed); (see 4.5.1) t_p = 300 ∝s; 2 percent maximum duty cycle	V_F	0.8	1.6	V_{pk}
Reverse current leakage	4016	DC method	I_{R1}			
1N5615, 1N5615UL, US		V_R = 200 V dc			0.5	∝A dc
1N5617, 1N5617UL, US		V_R = 400 V dc			0.5	∝A dc
1N5619, 1N5619UL, US		V_R = 600 V dc			0.5	∝A dc
1N5621, 1N5621UL, US		V_R = 800 V dc			0.5	∝A dc
1N5623, 1N5623UL, US		V_R = 1,000 V dc			0.5	∝A dc
Breakdown voltage	4021	I_R = 50 ∝A dc	V_{BR}			
1N5615, 1N5615UL, US				220		V dc
1N5617, 1N5617UL, US				440		V dc
1N5619, 1N5619UL, US				660		V dc
1N5621, 1N5621UL, US				880		V dc
1N5623, 1N5623UL, US				1100		V dc
Thermal impedance	3101	See 4.3.1 and 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation:		T_A = +100°C				
Reverse current leakage	4016	DC method	I_{R2}			
1N5615, 1N5615UL, US		V_R = 200 V dc			25	∝A dc
1N5617, 1N5617UL, US		V_R = 400 V dc			25	∝A dc
1N5619, 1N5619UL, US		V_R = 600 V dc			25	∝A dc
1N5621, 1N5621UL, US		V_R = 800 V dc			25	∝A dc
1N5623, 1N5623UL, US		V_R = 1,000 V dc			25	∝A dc

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Reverse recovery time	4031	Condition B1	t_{rr}			
1N5615, 1N5615UL, US					150	ns
1N5617, 1N5617UL, US					150	ns
1N5619, 1N5619UL, US					250	ns
1N5621, 1N5621UL, US					300	ns
1N5623, 1N5623UL, US					500	ns
Capacitance	4001	V_R = 12 V dc, 0.1 < f < 1.0 MHz	C			
1N5615, 1N5615UL, US					45	pF
1N5617, 1N5617UL, US					35	pF
1N5619, 1N5619UL, US					25	pF
1N5621, 1N5621UL, US					20	pF
1N5623, 1N5623UL, US					15	pF
Scope display evaluation	4023	See 4.5.4; n = 116, c = 0				
Subgroup 5						
Not applicable						
Subgroup 6						
Forward surge	4066	I_{FSM} = 25 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, I_O = 750 mA dc; V_R = rated V_{RWM} (see col. 3 of 1.3); T_A = +100°C; (T_{EC} = +100°C for surface mount devices (UL and US version)).				
Electrical measurement		See table I, subgroup 2 except thermal impedance.				
Subgroup 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
* Subgroup 1			45 devices c = 0
Thermal shock	1056	20 cycles, condition D except low temperature shall be achieved using liquid nitrogen (-195°C). Perform a visual for cracked glass.	
Temperature cycling	1051	500 cycles, -65°C to +175°C.	
Hermetic seal	1071		
Gross leak			
Electrical measurements		See table I, subgroup 2, and table III, steps 1 and 2.	
Subgroup 2			
Steady state dc blocking life	1048	1,000 hours, condition A $V_R = V_{RWM}$ (see col. 3 of 1.3).	
Electrical measurements		See table I, subgroup 2 herein except for thermal impedance.	22 devices c = 0
Subgroup 3			devices, c = 0
Decap analysis (DPA)	2101	Cross section and scribe and break. Separate samples shall be used for each test.	
Subgroup 4 1/			
Thermal impedance curves		Each supplier shall submit their (typical) maximum design thermal impedance curves. In addition, optional test conditions and $Z_{\theta JX}$ limit shall be provided to the qualifying activity in the qualification report.	
Subgroup 5			22 devices c = 0
Barometric pressure, reduced (altitude operation)	1001	Pressure 1N5617, 1N5619 = 8 mm Hg (100,000 ft); 1N5621, 1N5623 = 33 mm Hg (70,000 ft); $V_R = V_{RWM}$ (see col. 3 of 1.3); t = 1 minute (minimum).	

See footnote at end of table.

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* TABLE II. Group E inspection (all quality levels) for qualification and requalification only - Continued.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 7			
Not applicable			
Subgroup 8 1/			
Peak reverse power		See 4.5.6 and figure 6 herein. Peak reverse power (P_{RM}) shall be characterized by the supplier and this data shall be available to the government. Test shall be performed on each sublot.	
Electrical measurements		During the P_{RM} test, the voltage (V_{BR}) shall be monitored to verify it has not collapsed. Any collapse in V_{BR} during or after the P_{RM} test or rise in leakage current (I_R) after the test that exceeds I_{R1} in table I shall be considered a failure to that level of applied P_{RM}. Progressively higher levels of P_{RM} shall be applied until failure occurs on all devices within the chosen sample size to characterize each sublot.	
Subgroup 9 1/			45 devices c = 0
Resistance to glass cracking	1057	Step stress to destruction by increasing cycles or up to a maximum of 25 cycles.	
Subgroup 10			22 devices c = 0
Forward surge	4066	Condition A, I_{FSM} = rated (see col. 5 of 1.3.); 10 surges of 8.3 ms each at 1 minute intervals superimposed on I_O = 0; V_{RWM} = 0; T_A = + 25°C.	
Electrical measurement		See table I, subgroup 2.	

1/ The sample size for this step stress requirement shall be determined by the supplier. A statistically significant sample size is required.

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TABLE III. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current leaking change 1N5615, 1N5615UL, US 1N5617, 1N5617UL, US 1N5619, 1N5619UL, US 1N5621, 1N5621UL, US 1N5623, 1N5623UL, US	4016	DC method V_R = 200 V dc V_R = 400 V dc V_R = 600 V dc V_R = 800 V dc V_R = 1,000 V dc	ΔI_{R1} 4/		For JAN, JANTX, and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2.	Forward voltage change	4011	I_F = 3 A dc; Pulsed (see 4.5.1) t_p = 300 ∝s; 2 percent maximum duty cycle.	ΔV_{F1} 4/		± 50 mV dc maximum change from previous measured value.	

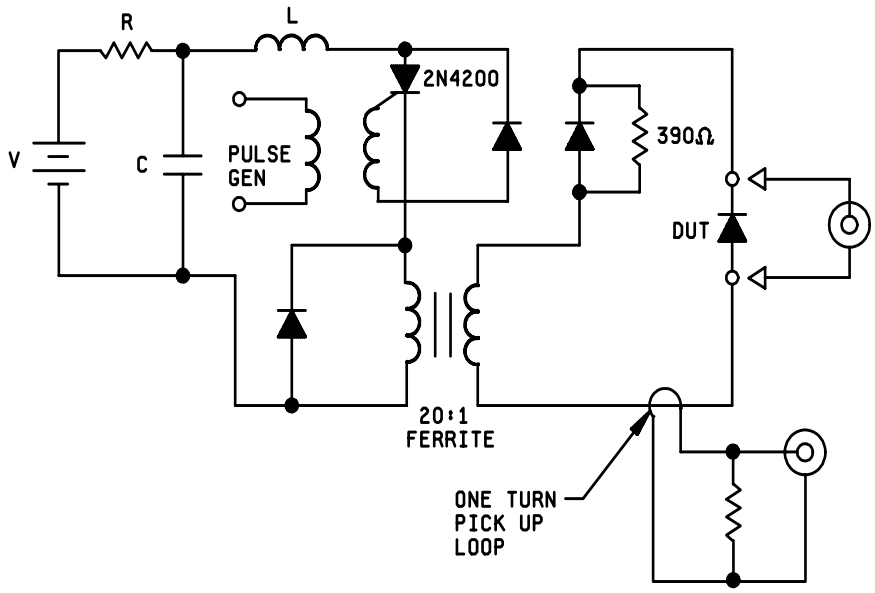
1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 2.
   b. Subgroup 4, see table III herein, step 2.
   c. Subgroup 5, see table III herein, step 1.

2/ The electrical measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 1.
   b. Subgroup 6, see table III herein, step 1.

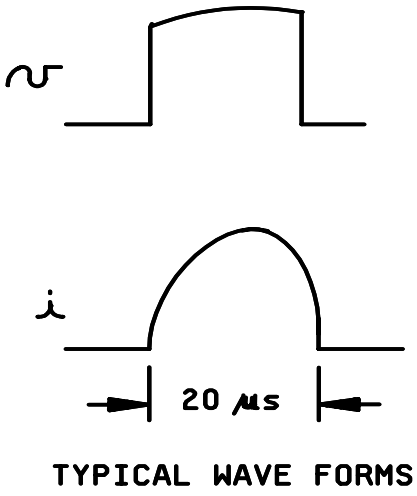
3/ The electrical measurements for table VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, step 1 (JANS).
   b. Subgroup 6, see table III herein, step 1 and 2 (JANS), step 1 (JAN, JANTX, and JANTXV).

4/ Devices which exceed the table I limits for this test shall not be accepted.

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NOTES: *
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 ∝fd to give a 20 ∝s pulse width.
3. V - adjustable to 200 volts for power desired in DUT.
   D1 - 3 kV; 600 Ma (1N3647 or equivalent).
   D2, D3 - 600 V; 3A (1N5552 or equivalent).
   * Values not stated are determined at the time of test.



TYPICAL WAVE FORMS

FIGURE 6. Peak reverse power measurement circuit and waveform.

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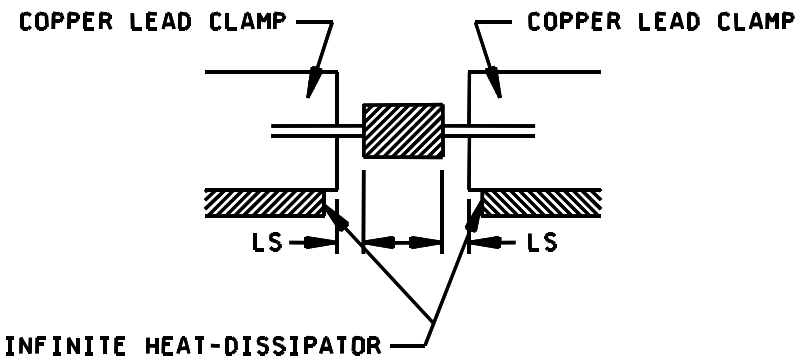


FIGURE 7. Mounting arrangement.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43216-5000 or e-mail vqe.chief@dla.mil.

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5615) will be identified on the Qualified Manufacturers List (QML).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5615	JANHCA1N5615 JANKCA1N5615	JANHCB1N5615 JANKCB1N5615
1N5617	JANHCA1N5617 JANKCA1N5617	JANHCB1N5617 JANKCB1N5617
1N5619	JANHCA1N5619 JANKCA1N5619	JANHCB1N5619 JANKCB1N5619
1N5621	JANHCA1N5621 JANKCA1N5621	JANHCB1N5621 JANKCB1N5621
1N5623	JANHCA1N5623 JANKCA1N5623	JANHCB1N5623 JANKCB1N5623

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6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 11
NASA - NA
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Review activities:
Army - AR, MI, SM
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Preparing activity:
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